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Introducción

Section 1

Utilites

Example of in-text citation of references in $\LaTeX$

The references in a report relate your content with the relevant sources, papers, and the works of others. To include references in a report, we *cite* them in the texts. In MS-Word, EndNote, or MS-Word references, or plain text as a list can be used. Similarly, in $\LaTeX$, you can use the “thebibliography” environment, which is similar to the plain text as a list arrangement like the MS word. However, In $\LaTeX$, the most convenient way is to use the BibTex, which takes the references in a particular format [see references.bib file of this template] and lists them in style [APA, Harvard, etc.] as we want with the help of proper packages.

These are the examples of how to *cite* external sources, seminal works, and research papers. In $\LaTeX$, if you use “**BibTex**” you do not have to worry much since the proper use of a bibliographystyle package like “agsm for the Harvard style” and little rectification of the content in a BiBText source file [In this template, BibTex are stored in the “references.bib” file], we can conveniently generate a reference style.

Take a note of the commands `\cite{}` and `\citep{}`. The command `\cite{}` will write like “Author et al. (2019)” style for Harvard, APA and Chicago style. The command `\citep{}` will write like “(Author et al., 2019).” Depending on how you construct a sentence, you need to use them smartly. Check the examples of **in-text citation** of sources listed here [This Department recommends the **Harvard style** of referencing.]:

- [Kottwitz \(2021\)](#) has written a comprehensive guide on writing in $\LaTeX$ [Example of `\cite{}`].
- If $\LaTeX$ is used efficiently and effectively, it helps in writing a very high-quality project report ([Lamport, 1994](#)) [Example of `\citep{}`].
- A detailed APA, Harvard, and Chicago referencing style guide are available in ([University of Reading, 2023](#)).

This is an example of how to construct a numbered list in $\LaTeX$, and it includes in-text, named and parenthetical citations:

1. [Kottwitz \(2021\)](#) has written a comprehensive guide on writing in $\LaTeX$.
2. If $\LaTeX$ is used efficiently and effectively, it helps in writing a very high-quality project report ([Lamport, 1994](#)).

Reference Resources

You can find additional referencing resources from the University Library:

- <https://libguides.reading.ac.uk/computer-science>
- <https://libguides.reading.ac.uk/citing-references/citationexamples>

Changing Bibliography Styles

While this report used name and date formatting in the Harvard style, you might also wish to use a numbered style like that from IEEE. To enable this change, you will need to edit the `CS_report.sty` file. Uncomment the 2 lines of Harvard settings under Bibliography/References settings, and enable the IEEE style:

```
% IEEE, Numbered Style
\usepackage[numbers]{natbib}
\bibliographystyle{IEEEtran}
```

Note that when making this change, you will need to modify the way in which you refer to authors by name, as this is no longer immediately automatic. Instead, you will need to additionally rely on `\citeauthor{}`.

Example of an Equation in $\LaTeX$

Eq. 1 [note that this is an example of an equation's in-text citation] is an example of an equation in $\LaTeX$. In Eq. (1), s is the mean of elements $x_i \in \mathbf{x}$:

$$s = \frac{1}{N} \sum_{i=1}^N x_i. \quad (1)$$

Have you noticed that all the variables of the equation are defined using the **in-text** maths command `$. $`, and Eq. (1) is treated as a part of the sentence with proper punctuation? Always treat an equation or expression as a part of the sentence.

Example of an algorithm in $\LaTeX$

Algorithm 1 is a good example of an algorithm in $\LaTeX$.

Example of code snippet in $\LaTeX$

Code Listing 1 is a good example of including a code snippet in a report. While using code snippets, take care of the following:

- do not paste your entire code (implementation) or everything you have coded. Add code snippets only.

Algorithm 1 Example caption: sum of all even numbers

Require: $\mathbf{x} = x_1, x_2, \dots, x_N$

Ensure: *EvenSum* (Sum of even numbers in $\mathbf{x}$)

```
1: function EvenSummation( $\mathbf{x}$ )
2:    $EvenSum \leftarrow 0$ 
3:    $N \leftarrow \text{length}(\mathbf{x})$ 
4:   for  $i \leftarrow 1$  to  $N$  do
5:     if  $x_i \bmod 2 == 0$  then                                ▷ Check whether a number is even.
6:        $EvenSum \leftarrow EvenSum + x_i$ 
7:     end if
8:   end for
9:   return  $EvenSum$ 
10: end function
```

- The algorithm shown in Algorithm 1 is usually preferred over code snippets in a technical/scientific report.
- Make sure the entire code snippet or algorithm stays on a single page and does not overflow to another page(s).

Here are three examples of code snippets for three different languages (Python, Java, and CPP) illustrated in Listings 1, 2, and 3 respectively.

```
1 import numpy as np
2
3 x = [0, 1, 2, 3, 4, 5] # assign values to an array
4 evenSum = evenSummation(x) # call a function
5
6 def evenSummation(x):
7     evenSum = 0
8     n = len(x)
9     for i in range(n):
10         if np.mod(x[i],2) == 0: # check if a number is even?
11             evenSum = evenSum + x[i]
12     return evenSum
```

Listing 1: Code snippet in $\text{\LaTeX}$ and this is a Python code example

Here we used the “\clearpage” command and forced-out the second listing example onto the next page.

```

1 public class EvenSum{
2     public static int evenSummation(int[] x){
3         int evenSum = 0;
4         int n = x.length;
5         for(int i = 0; i < n; i++){
6             if(x[i]%2 == 0){ // check if a number is even?
7                 evenSum = evenSum + x[i];
8             }
9         }
10        return evenSum;
11    }
12    public static void main(String[] args){
13        int[] x = {0, 1, 2, 3, 4, 5}; // assign values to an array
14        int evenSum = evenSummation(x);
15        System.out.println(evenSum);
16    }
17 }

```

Listing 2: Code snippet in $\LaTeX$ and this is a Java code example

```

1 int evenSummation(int x[]){
2     int evenSum = 0;
3     int n = sizeof(x);
4     for(int i = 0; i < n; i++){
5         if(x[i]%2 == 0){ // check if a number is even?
6             evenSum = evenSum + x[i];
7         }
8     }
9     return evenSum;
10 }
11
12 int main(){
13     int x[] = {0, 1, 2, 3, 4, 5}; // assign values to an array
14     int evenSum = evenSummation(x);
15     cout<<evenSum;
16     return 0;
17 }

```

Listing 3: Code snippet in $\LaTeX$ and this is a C/C++ code example

Conclusión

El desarrollo del sistema...

Bibliografía

Kottwitz, S. (2021), *LaTeX beginner's guide : create visually appealing texts, articles, and books for business and science using LaTeX*, Packt. ISBN: 9781801072588.

Lamport, L. (1994), *LATEX: a document preparation system: user's guide and reference manual*, Addison-wesley.

University of Reading (2023), 'Different styles & systems of referencing: Guidance on citing references for students at the university of reading'. (accessed June 6, 2023).
URL: <https://libguides.reading.ac.uk/citing-references/referencingstyles>